AI Intern Assessment: Real-Time Static Hand Gesture Recognition.

Submission Deadline: 02/09/2025 by 5pm IST.
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1. Objective

The primary goal of this assignment is to design and implement a Python application that uses a webcam to recognize a predefined set of static hand gestures in real-time. This task will assess your ability to work with live video data, select and implement appropriate computer vision tools, and apply logical reasoning to interpret visual information.

2. Background

Hand gesture recognition is a fundamental problem in Human-Computer Interaction (HCI). A robust solution requires a multi-stage process: detecting a hand, extracting key features or landmarks, and classifying the gesture. Your challenge is to research and implement a solution to this problem, making your own informed decisions about the best technical approach.

3. Detailed Task Description

You are required to develop a Python application that performs the following steps:

- **Environment Setup:** Create a structured project environment that is easy for others to replicate.
- **Webcam Integration:** The application must capture a live video feed from a standard webcam as its primary input.
- **Hand Detection & Tracking:** Implement or integrate a method to detect and track the position and features of a hand within the video frames. You are free to choose the technology you believe is best suited for this task.
- **Gesture Logic Implementation:** Develop a rules-based or model-based algorithm to classify the hand's pose into one of the required gestures. This logic should be based on the features you extract from the detected hand.
- **Real-Time Display:** The application must display the webcam feed in a window. It should visually indicate where the hand is detected (e.g., by drawing a bounding box or keypoints) and, most importantly, overlay the name of the currently recognized gesture as text on the screen.

4. Core Requirements

Your final submission must meet the following criteria:

- **Gesture Vocabulary:** The application must successfully recognize a minimum of four distinct static gestures:
    a. Open Palm
    b. Fist
    c. Peace Sign (V-sign)
    d. Thumbs Up
- **Technology Choice:** You must select and use a computer vision library, framework, or technique appropriate for the task. Your choice must be justified in your project's documentation.
- **Performance:** The recognition must happen in real-time on a live video stream without significant lag.
- **User Feedback:** The name of the recognized gesture (e.g., "Fist") must be clearly displayed on the output video window.
- **Code Quality:** The Python code must be clean, well-organized, and include comments where necessary to explain complex parts of your logic.

5. Deliverables

You are required to create a public Git repository (e.g., on GitHub, GitLab) and submit the link to it. The repository must be professionally structured and contain the following:

- **Source Code:** All Python scripts and related files required to run the project.
- **Dependency Manifest:** A requirements.txt file listing all necessary Python libraries.
- **README.md:** A detailed README file is crucial for this assignment. It must include:
- Your full name.
- **Technology Justification:** A section explaining the libraries/frameworks you chose for hand detection and why you believe they were the best choice for this specific problem.
- **Gesture Logic Explanation:** A clear description of the methodology you used to identify each of the four required gestures.
- **Setup and Execution Instructions:** Clear, step-by-step instructions on how to set up the environment and run your application.
- **Demonstration:** A short screen recording (demo.gif or demo.mp4) showing your application successfully identifying the four required gestures. You may either include this file in the repository or provide a link to it in your README.